

SECoP@HMC – Data Storage

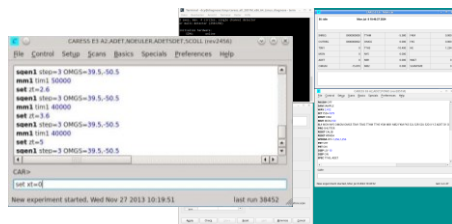
Making sample environment (meta)data understandable



Sample Environment Communication Protocol

HZB

Experiment Control

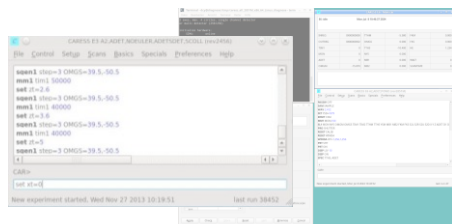


Sample Environment

Sample Environment Communication Protocol

HZB

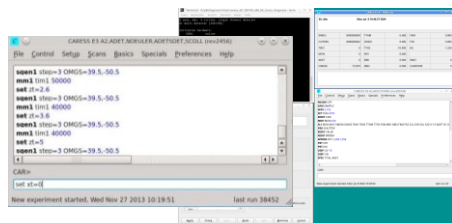
Experiment Control



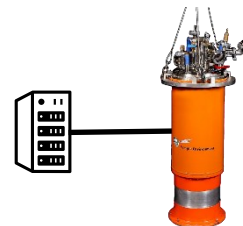
Sample Environment

HZB

Experiment Control
+ SECoP Client



SECoP

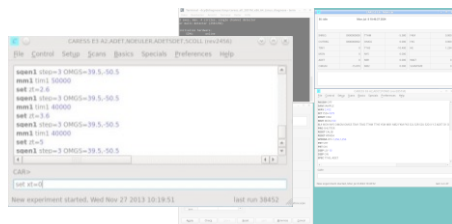


Sample Environment
+ SECoP Server (SENode)

Sample Environment Communication Protocol

HZB

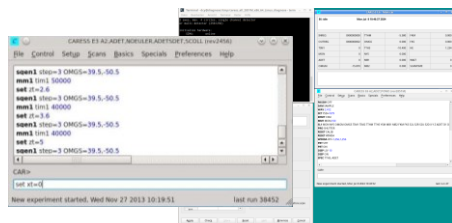
Experiment Control



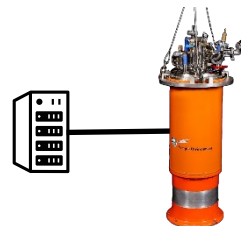
Sample Environment

HZB

Experiment Control
+ SECoP Client



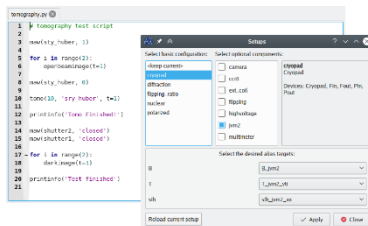
SECoP



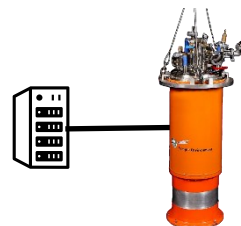
Sample Environment
+ SECoP Server (SENode)

MLZ
Heinz Maier-Leibnitz Zentrum

Experiment Control
+ SECoP Client



SECoP



Sample Environment
+ SECoP Server (SENode)



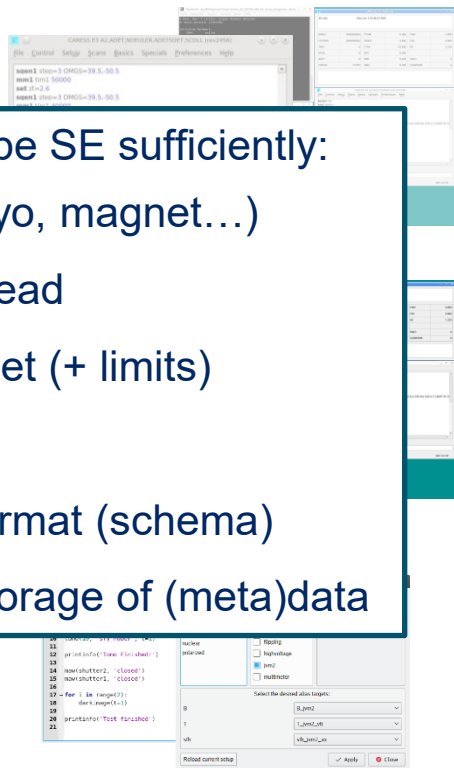
Sample Environment Communication Protocol

HZB

Metadata to describe SE sufficiently:

- Model / kind (cryo, magnet...)
 - Parameters to read
 - Parameters to set (+ limits)
 - Units
- ⇒ Standardised format (schema)
- ⇒ Standardised storage of (meta)data

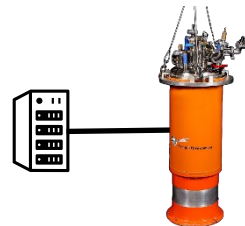
Experiment Control
+ SECoP Client



Sample Environment



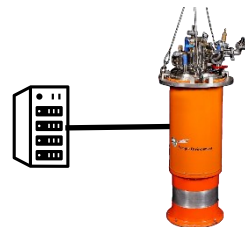
SECoP



Sample Environment
+ SECoP Server (SENode)



SECoP



Sample Environment
+ SECoP Server (SENode)



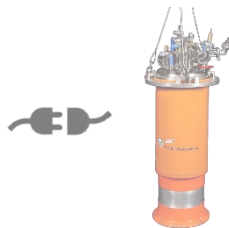
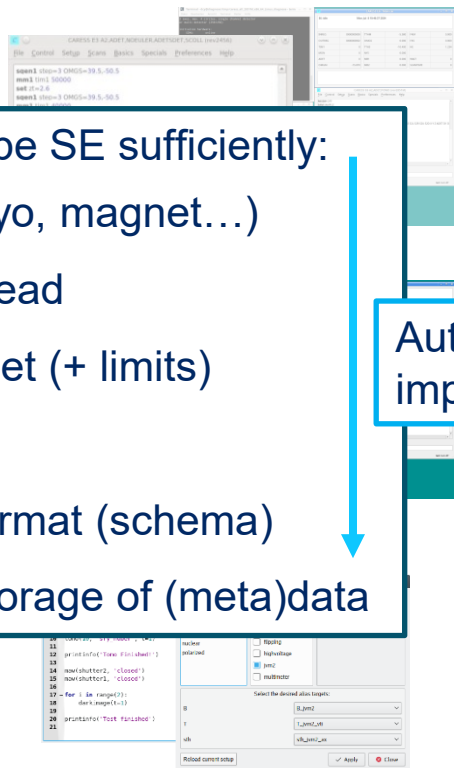
Sample Environment Communication Protocol

HZB

Metadata to describe SE sufficiently:

- Model / kind (cryo, magnet...)
 - Parameters to read
 - Parameters to set (+ limits)
 - Units
- ⇒ Standardised format (schema)
- ⇒ Standardised storage of (meta)data

Experiment Control
+ SECoP Client



Sample Environment

Automatic storage of
important metadata

Sample Environment
+ SECoP Server (SENode)

SECoP



Sample Environment
+ SECoP Server (SENode)

Sample Environment Communication Protocol

```
> describe
< describing .
{
  "description":"TestNode",
  "equipment_id":"HZB_Testnode1",
  "firmware":"SHALL server library (Git 70591a14f66f37b92dcf6386a17159b526fa2913)",
  "modules":
  {
    "temp1":
    {
      "interface_classes":["Writable","Readable"],
      "description":"a meaningful description of the module",
      "accessibles":
      {
        "value":
        {
          "description":"temperature",
          "datainfo":{"type":"double","unit":"K"},
          "readonly":true
        },
        "target":
        {
          "description":"target temperature",
          "datainfo":{"type":"double","unit":"K"},
          "readonly":false
        }
      }
    }
  }
}
```

General description (static parameters)

List of modules (dynamic parameters)
(e.g. a magnet could read/set several temperatures and a magnetic field)

Experiment control can read
current temperature

Experiment control can set
target temperature

SECoP: Meaning

Each module defines what it does:

"meaning": {

"key": "sample temperature",

(human readable)

"link": "http://purl.allotrope.org/ontologies/result#AFR_0002149 ",

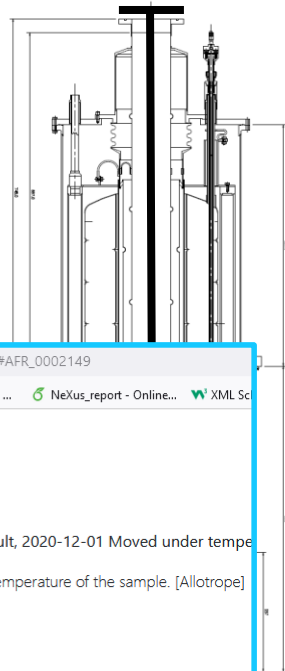
(machine readable)

"function": "temperature_regulation",

(SECoP class)

"importance": 20,

"belongs_to": "sample" }



← → ↻ 🏠 purl.allotrope.org/ontologies/result#AFR_0002149

Perplexity How to Install a Linux ... Meeting eröffnen - Zo... Gerit Guenther / My ... NeXus_report - Online... XML Sc

sample temperature

(http://purl.allotrope.org/ontologies/result#AFR_0002149)

synonyms: 2020-12-01 Changed labels. [Allotrope], sample temperature result, 2020-12-01 Moved under tempe

A sample temperature result is a quality quantification facet that quantifies the temperature of the sample. [Allotrope]

© Allotrope

Creative Commons Attribution 4.0 International Public License

defined in:

- <http://purl.allotrope.org/voc/afo/REC/2020/12/aft>
- <http://purl.allotrope.org/voc/afo/REC/2021/03/aft>

SECoP: Meaning

Each module defines what it does:

```
"meaning": {
```

```
  "key": "sample temperature",
```

(human readable)

```
  "link": "http://purl.allotrope.org/ontologies/result#AFR_0002149 ",
```

(machine readable)

```
  "function": "temperature_regulation",
```

(SECoP class)

```
  "importance": 20,
```

(artificial parameter)

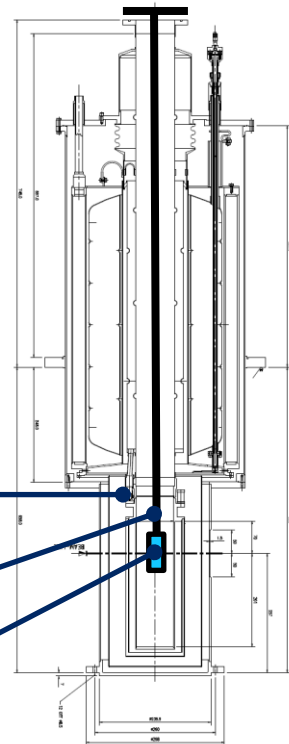
```
  "belongs_to": "sample" }
```

(applies to)

SE temperature
(importance 5)

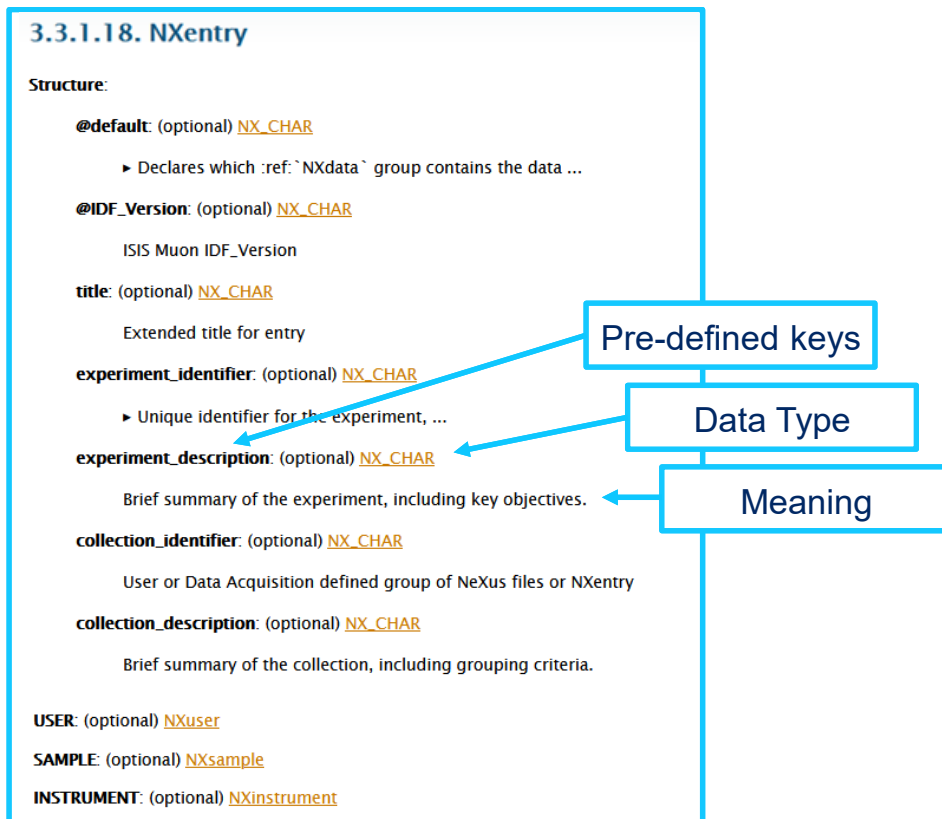
Sample stick temperature
(importance 10)

Sample holder temperature
(importance 20)



NeXus

- Binary file format (HDF5)
- Hierarchical, defined structure
 - Groups
 - Data Fields
 - @Attributes
- Metadata + data create context
- Persistent web description (URL)
- Modular structure (base classes)
- Metadata schema for experimental data



NeXus

- Binary file format (HDF5)
- Hierarchical, defined structure
 - Groups
 - Data Fields
 - @Attributes
- Metadata + data create context
- Persistent web description (URL)
- Modular structure (base classes)
- Metadata schema for experimental data: logic through structure & semantics

3.3.1.18. NXentry

Structure:

@default: (optional) [NX_CHAR](#)

► Declares which :ref:`NXdata` group

@IDF_Version: (optional) [NX_CHAR](#)

ISIS Muon IDF_Version

title: (optional) [NX_CHAR](#)

Extended title for entry

experiment_identifier: (optional) [NX_CHAR](#)

► Unique identifier for the experiment, ...

experiment_description: (optional) [NX_CHAR](#)

Brief summary of the experiment, inc

collection_identifier: (optional) [NX_CHAR](#)

User or Data Acquisition defined gro

collection_description: (optional) [NX_CHAR](#)

Brief summary of the collection, inclu

USER: (optional) [NXuser](#)

SAMPLE: (optional) [NXsample](#)

INSTRUMENT: (optional) [NXinstrument](#)

3.3.1.56. NXuser

Structure:

@default: (optional) [NX_CHAR](#)

► Declares which child group contains a path leading .

name: (optional) [NX_CHAR](#)

Name of user responsible for this entry

role: (optional) [NX_CHAR](#)

► Role of user responsible for this entry. ...

3.3.1.47. NXsample

Structure:

@default: (optional) [NX_CHAR](#)

► Declares which child group contains a path leading ...

name: (optional) [NX_CHAR](#)

Descriptive name of sample

chemical_formula: (optional) [NX_CHAR](#)

► The chemical formula specified using CIF conventions. ...

```
{
  "description": "TestNode",
  "equipment_id": "HZB_Testnode1",
  "firmware": "SHALL server library (Git 70591a14f6",
  "modules":
  {
    "temp1":
    {
      "interface_classes": ["Writable", "Readable"],
      "description": "a meaningful description of t",
      "accessibles":
      {
        "value":
        {
          "description": "temperature",
          "datainfo": {"type": "double", "unit": "K"},
          "readonly": true
        },
      },
      "meaning":
      {
        "key": "sample temperature",
        "link": "http://purl.allotrope.org/ontologies/result#AFR_0002149 ",
        "function": "temperature_regulation",
        "importance": 20,
        "belongs_to": "sample"
      },
    },
  },
}
```

3.3.1.47. NXsample

Structure:

name: (optional) [NX_CHAR](#)

Descriptive name of sample

temperature: (optional) [NX_FLOAT](#) (Rank: anyRank, Dimensions: [1

Sample temperature. This could be a scanned variable

temperature_env: (optional) [NXenvironment](#)

Additional sample temperature environment information

magnetic_field: (optional) [NXlog](#)

magnetic_field.value is a link to e.g. magnetic_field_env.sensor1.value

chemical_formula: (optional) [NX_CHAR](#)

► The chemical formula specified using CIF conventions. ...

```
{
  "description": "TestNode",
  "equipment_id": "HZB_Testnode1",
  "firmware": "SHALL server library (Git 70591a14f6)",
  "modules": {
    "temp1": {
      "interface_classes": ["Writable", "Readable"],
      "description": "a meaningful description of temperature",
      "accessibles": {
        "value": {
          "description": "temperature",
          "datainfo": { "type": "double", "unit": "K" },
          "readonly": true
        }
      },
      "meaning": {
        "key": "sample temperature",
        "link": "http://purl.allotrope.org/ontologies/result#AFR_0002149 ",
        "function": "temperature_regulation",
        "importance": 20,
        "belongs_to": "sample"
      }
    }
  }
}
```

3.3.1.47. NXsample

Structure:

name: (optional) [NX_CHAR](#)

Descriptive name of sample

temperature: (optional) [NX_FLOAT](#) (Rank: anyRank)

Sample temperature. This could be a scanned

temperature_env: (optional) [NXenvironment](#)

Additional sample temperature environment info

magnetic_field: (optional) [NXlog](#)

magnetic_field.value is a link to e.g. magnetic_field_env.sensor1.value

chemical_formula: (optional) [NX_CHAR](#)

► The chemical formula specified using CIF conventions. ...

3.3.1.19. NXenvironment

Structure:

name: (optional) [NX_CHAR](#)

Apparatus identification code/model number; e.g. OC100 011

description: (optional) [NX_CHAR](#)

Description of the apparatus; e.g. 100mm bore orange cryostat with Roots pump

SENSOR: (optional) [NXsensor](#)

SECoP ⇒ NeXus

```
{
  "description": "TestNode",
  "equipment_id": "HZB_Testnode1",
  "firmware": "SHALL server library (Git 70591a14f6)",
  "modules": {
    "temp1": {
      "interface_classes": ["Writable", "Readable"],
      "description": "a meaningful description of the module",
      "accessibles": {
        "value": {
          "description": "temperature",
          "datainfo": {"type": "double", "unit": "K"},
          "readonly": true
        },
        "meaning": {
          "key": "sample temperature",
          "link": "http://purl.allotrope.org/ontologies/result#AFR_0002149",
          "function": "temperature_regulation",
          "importance": 20,
          "belongs_to": "sample"
        }
      }
    }
  }
}
```

3.3.1.47. NXsample

Structure:

- name:** (optional) [NX_CHAR](#)
Descriptive name of sample
- temperature:** (optional) [NX_FLOAT](#) (Rank: anyRank)
Sample temperature. This could be a scanned value.
- temperature_env:** (optional) [NXenvironment](#)
Additional sample temperature environment information.
- magnetic_field:** (optional) [NXlog](#)
magnetic_field.value is a log of the magnetic field.
- chemical_formula:** (optional) [NX_CHAR](#)
The chemical formula of the sample.

3.3.1.19. NXenvironment

Structure:

- name:** (optional) [NX_CHAR](#)
Apparatus identification code/model number; e.g. OC100 011
- description:** (optional) [NX_CHAR](#)
Description of the apparatus; e.g. 100mm bore orange cryostat with Roots pump
- SENSOR:** (optional) [NXsensor](#)

3.3.1.49. NXsensor

Structure:

- name:** (optional) [NX_CHAR](#)
Name for the sensor
- value_log:** (optional) [NXlog](#)
Time history of sensor readings
- measurement:** (optional) [NX_CHAR](#)
name for measured signal ...
Any of these values:
 - [temperature](#)
 - [pH](#)
 - [magnetic_field](#)
 - [electric_field](#)
 - [conductivity](#)
 - [resistance](#)
 - [voltage](#)

SECoP ⇒ NeXus

```
{
  "description": "TestNode",
  "equipment_id": "HZB_Testnode1",
  "firmware": "SHALL server library (Git 70591a14f6)",
  "modules": {
    "temp1": {
      "interface_classes": ["Writable", "Readable"],
      "description": "a meaningful description of the module",
      "accessibles": {
        "value": {
          "description": "temperature",
          "datainfo": {
            "type": "double",
            "unit": "K",
            "readonly": true
          },
          "meaning": {
            "key": "sample temperature",
            "link": "http://purl.allotrope.org/ontologies/result#AFR_0002149",
            "function": "temperature_regulation",
            "importance": 20,
            "belongs_to": "sample"
          }
        }
      }
    }
  }
}
```

3.3.1.47. NXsample

Structure:

- name:** (optional) [NX_CHAR](#)
Descriptive name of sample
- temperature:** (optional) [NX_FLOAT](#) (Rank: anyRank)
Sample temperature. This could be a scanned value
- temperature_env:** (optional) [NXenvironment](#)
Additional sample temperature environment information
- magnetic_field:** (optional) [NXlog](#)
magnetic_field.value is an array of [NXlog](#) objects
- chemical_formula:** (optional) [NX_CHAR](#)
The chemical formula of the sample

3.3.1.19. NXenvironment

Structure:

- name:** (optional) [NX_CHAR](#)
Apparatus identification code/model number; e.g. OC100 011
- description:** (optional) [NX_CHAR](#)
Description of the apparatus; e.g. 100mm bore orange cryostat with Roots pump
- SENSOR:** (optional) [NXsensor](#)

3.3.1.49. NXsensor

Structure:

- name:** (optional) [NX_CHAR](#)
Name for the sensor
- value_log:** (optional) [NXlog](#)
Time history of sensor reading
- measurement:** (optional) [NX_CHAR](#)
name for measured signal ...
Any of these values:
 - [temperature](#)
 - [pH](#)
 - [magnetic_field](#)
 - [electric_field](#)
 - [conductivity](#)
 - [resistance](#)
 - [voltage](#)

3.3.1.30. NXlog

Structure:

- time:** (optional) [NX_NUMBER](#) {units=[NX_TIME](#)}
Time of logged entry. The times are relative to the "start" attribute ...
- @start:** (optional) [NX_DATE_TIME](#)
- @scaling_factor:** (optional) [NX_NUMBER](#)
- value:** (optional) [NX_NUMBER](#) {units=[NX_ANY](#)}
Array of logged value, such as temperature. If this is ...

SECoP ⇒ NeXus

```
{
  "description": "TestNode",
  "equipment_id": "HZB_Testnode1",
  "firmware": "SHALL server library (Git 70591a14f6)",
  "modules": {
    "temp1": {
      "interface_classes": ["Writable", "Readable"],
      "description": "a meaningful description of the module",
      "accessibles": {
        "value": {
          "description": "temperature",
          "datainfo": {
            "type": "double",
            "unit": "K",
            "readonly": true
          },
          "meaning": {
            "key": "sample temperature",
            "link": "http://purl.allotrope.org/ontologies/result#AFR_0002149",
            "function": "temperature_regulation",
            "importance": 20,
            "belongs_to": "sample"
          }
        }
      }
    }
  }
}
```

3.3.1.47. NXsample

Structure:

- name:** (optional) [NX_CHAR](#)
Descriptive name of sample
- temperature:** (optional) [NX_FLOAT](#) (Rank: anyRank)
Sample temperature. This could be a scanned value
- temperature_env:** (optional) [NXenvironment](#)
Additional sample temperature environment information
- magnetic_field:** (optional) [NXlog](#)
magnetic_field.value is an [NXlog](#)
- chemical_formula:** (optional) [NX_CHAR](#)
The chemical formula

3.3.1.19. NXenvironment

Structure:

- name:** (optional) [NX_CHAR](#)
Apparatus identification code/model number; e.g. OC100 011
- description:** (optional) [NX_CHAR](#)
Description of the apparatus; e.g. 100mm bore orange cryostat with Roots pump
- SENSOR:** (optional) [NXsensor](#)

3.3.1.49. NXsensor

Structure:

- name:** (optional) [NX_CHAR](#)
Name for the sensor
- value_log:** (optional) [NXlog](#)
Time history of sensor reading
- measurement:** (optional) [NX_CHAR](#)
name for measured signal ...
Any of these values:
 - [temperature](#)
 - [pH](#)
 - [magnetic_field](#)
 - [electric_field](#)
 - [conductivity](#)
 - [resistance](#)
 - [voltage](#)

3.3.1.30. NXlog

Structure:

- time:** (optional) [NX_NUMBER](#) {units=[NX_TIME](#)}
Time of logged entry. The times are relative to the "start" attribute ...
- @start:** (optional) [NX_DATE_TIME](#)
- @scaling_factor:** (optional) [NX_NUMBER](#)
- value:** (optional) [NX_NUMBER](#) {units=[NX_ANY](#)}
Array of logged value, such as temperature. If this is ...

SECoP ⇒ NeXus

```
{  
  "description": "TestNode",  
  "equipment_id": "HZB_Testnode1",  
  "firmware": "SHALL server library (Git 70591a14f6)",  
  "modules":  
  {  
    "temp1":  
    {  
      "interface_classes": ["Writable", "Readable"],  
      "description": "a meaningful description of the module",  
      "accessibles":  
      {  
        "value":  
        {  
          "description": "temperature",  
          "datainfo": {"type": "double", "unit": "K"},  
          "readonly": true  
        },  
        "meaning":  
        {  
          "key": "sample temperature",  
          "link": "http://purl.allotrope.org/ontologies/result#AFR_0002145",  
          "function": "temperature_regulation",  
          "importance": 20,  
          "belongs_to": "sample"  
        }  
      }  
    }  
  }  
}
```

3.3.1.47. NXsample

Structure:

name: (optional) [NX_CHAR](#)
Descriptive name of sample

temperature: (optional) [NX_FLOAT](#) (Rank: anyRank)
Sample temperature. This could be a scanned value.

temperature_env: (optional) [NXenvironment](#)
Additional sample temperature environment information.

magnetic_field: (optional) [NXlog](#)
magnetic_field.value is an NXlog.

chemical_formula: (optional) [NX_CHAR](#)
The chemical formula.

3.3.1.19. NXenvironment

Structure:

name: (optional) [NX_CHAR](#)
Apparatus identification code/model number; e.g. OC100 011

description: (optional) [NX_CHAR](#)
Description of the apparatus; e.g. 100mm bore orange cryostat with Roots pump

SENSOR: (optional) [NXsensor](#)

3.3.1.49. NXsensor

Structure:

name: (optional) [NX_CHAR](#)
Name for the sensor

value_log: (optional) [NXlog](#)
Time history of sensor readings.

measurement: (optional) [NX_CHAR](#)
name for measured signal ...
Any of these values:

- [temperature](#)
- [pH](#)
- [magnetic_field](#)
- [electric_field](#)
- [conductivity](#)
- [resistance](#)
- [voltage](#)

3.3.1.30. NXlog

Structure:

time: (optional) [NX_NUMBER](#) {units=[NX_TIME](#)}
Time of logged entry. The times are relative to the "start" attribute ...

@start: (optional) [NX_DATE_TIME](#)

@scaling_factor: (optional) [NX_NUMBER](#)

value: (optional) [NX_NUMBER](#) {units=[NX_ANY](#)}
Array of logged value, such as temperature. If this is ...

SECoP ⇒ NeXus

```
{
  "description": "TestNode",
  "equipment_id": "HZB_Testnode1",
  "firmware": "SHALL server library (Git 70591a14f6)",
  "modules": {
    "temp1": Module
  }
  {
    "interface_classes": ["Writable", "Readable"],
    "description": "a meaningful description of the module",
    "accessibles": {
      "value": Parameter
    }
    {
      "description": "temperature",
      "datainfo": { "type": "double", "unit": "K" },
      "readonly": true
    },
    "meaning": {
      "key": "sample temperature",
      "link": "http://purl.allotrope.org/ontologies/result#AFR_0002145",
      "function": "temperature_regulation",
      "importance": 20,
      "belongs_to": "sample"
    }
  },
}
```

3.3.1.47. NXsample

Structure:

name: (optional) [NX_CHAR](#)
Descriptive name of sample

temperature: (optional) [NX_FLOAT](#) (Rank: anyRank)
Sample temperature. This could be a scanned value.

temperature_env: (optional) [NXenvironment](#)
Additional sample temperature environment information.

magnetic_field: (optional) [NXlog](#)
magnetic_field.value is an array of [NXlog](#) objects.

chemical_formula: (optional) [NX_CHAR](#)
The chemical formula of the sample.

3.3.1.19. NXenvironment

Structure:

name: (optional) [NX_CHAR](#)
Apparatus identification code/model number; e.g. OC100 011

description: (optional) [NX_CHAR](#)
Description of the apparatus; e.g. 100mm bore orange cryostat with Roots pump

SENSOR: (optional) [NXsensor](#)

3.3.1.49. NXsensor

Structure:

name: (optional) [NX_CHAR](#)
Name for the sensor

value_log: (optional) [NXlog](#)
Time history of sensor readings.

measurement: (optional) [NX_CHAR](#)
name for measured signal ...
Any of these values:

- temperature
- pH
- magnetic_field
- electric_field
- conductivity
- resistance
- voltage

3.3.1.30. NXlog

Structure:

time: (optional) [NX_NUMBER](#) {units=[NX_TIME](#)}
Time of logged entry. The times are relative to the "start" attribute ...

@start: (optional) [NX_DATE_TIME](#)

@scaling_factor: (optional) [NX_NUMBER](#)

value: (optional) [NX_NUMBER](#) {units=[NX_ANY](#)}
Array of logged value, such as temperature. If this is ...

3.3.1.8. NXcollection

Description:

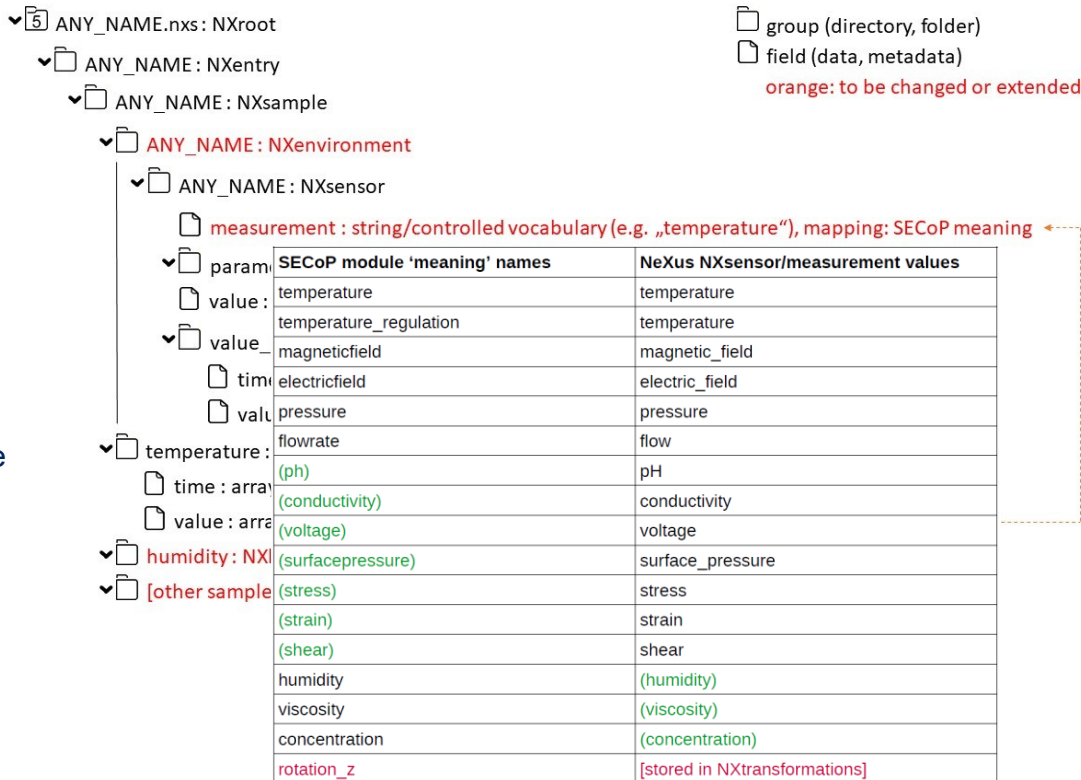
► An unvalidated set of terms, such as the description of a beam line. ...

Everything else

SECoP ⇒ NeXus

Required modifications:

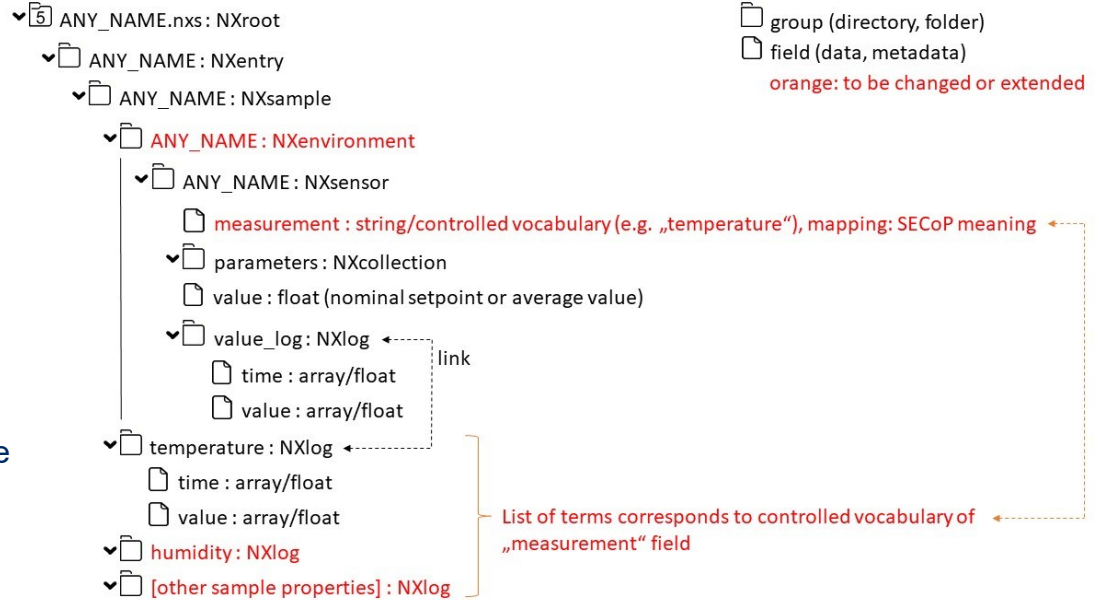
- Extension of 'measurement' field ('humidity', 'viscosity', and 'concentration')
- Add corresponding fields to NXsample
- Adding 'NXenvironments' according to 'measurement' fields (currently only temperature_env and magnetic_field_env are allowed)
- Allow any 'NXenvironment' (ANY_NAME : NXenvironment)



SECoP ⇒ NeXus

Required modifications:

- Extension of ‚measurement‘ field ('humidity', 'viscosity', and 'concentration')
- Add corresponding fields to NXsample
- Adding ‚NXenvironments‘ according to ‚measurement‘ fields (currently only temperature_env and magnetic_field_env are allowed)
- Allow any ‚NXenvironment‘ (ANY_NAME : NXenvironment)



SECoP ⇒ NeXus: Implementation

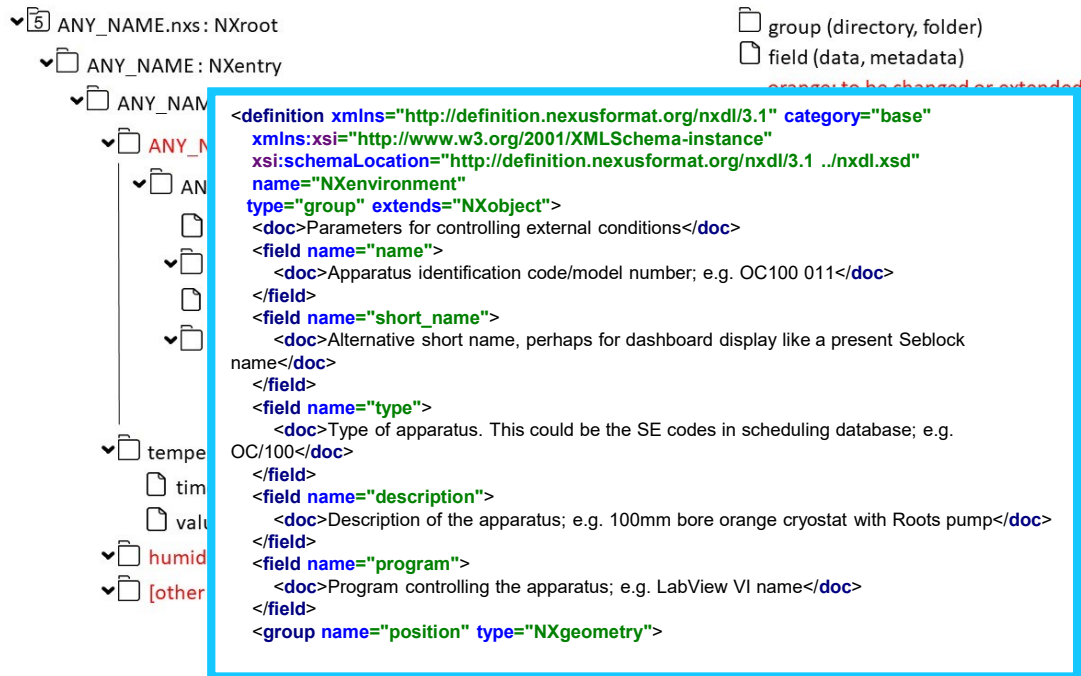
How to implement changes?

1. Extension of NeXus standard:

- NXDL description of changes
- Proposal to NIAC through representatives

2. Meanwhile:

- Make modified NXDL available via PID (e.g. at Zenodo)
- Refer from NeXus field 'definition_local' to modified NXDL



SECoP $\Rightarrow$ NeXus: Implementation

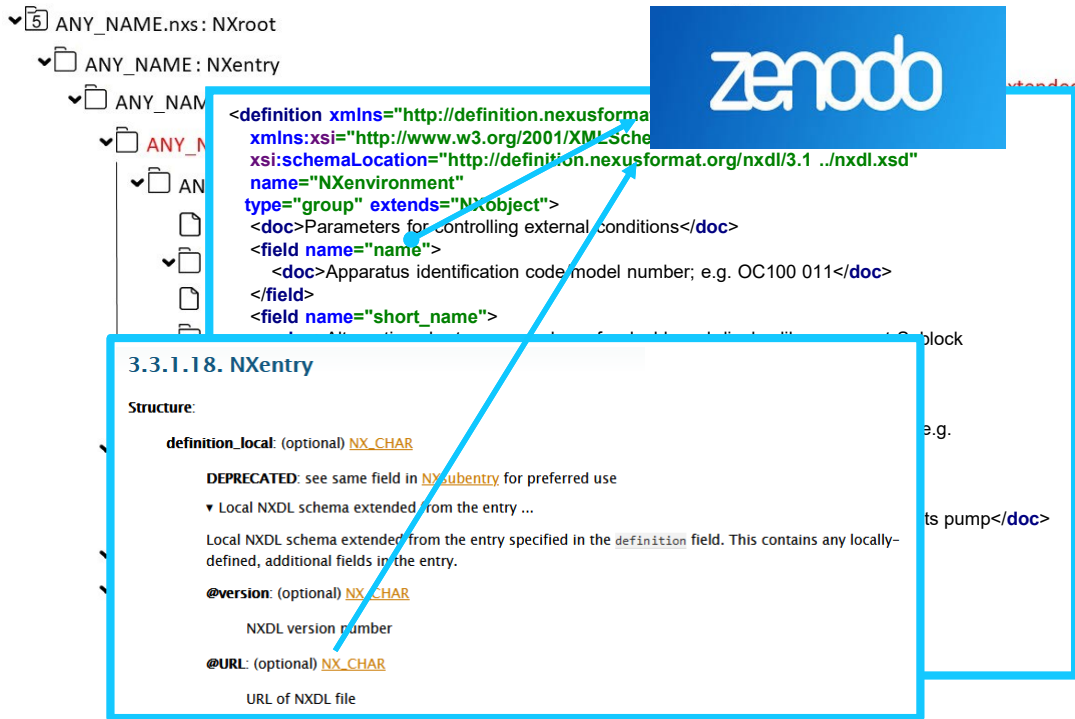
How to implement changes?

1. Extension of NeXus standard:

- NXDL description of changes
- Proposal to NIAC through representatives

2. Meanwhile:

- Make modified NXDL available via PID (e.g. at Zenodo)
- Refer from NeXus field 'definition_local' to modified NXDL



SECoP: Meaning

Each module defines what it does:

"meaning": {

 "key": "sample temperature",

 "link": "http://purl.allotrope.org/ontologies/result#AFR_0002149 ",

 "function": "temperature_regulation",

 "importance": 20,

 "belongs_to": "sample" }

